

EXECUTIVE BRIEFING · ARCHITECTURE OVERVIEW

Cyber Threat Analytics Dashboard

Transforming fragmented incident data into a single, predictive operational intelligence engine.

850 raw incident records unified

8 reporting sources standardized

90 days of cross-filterable activity

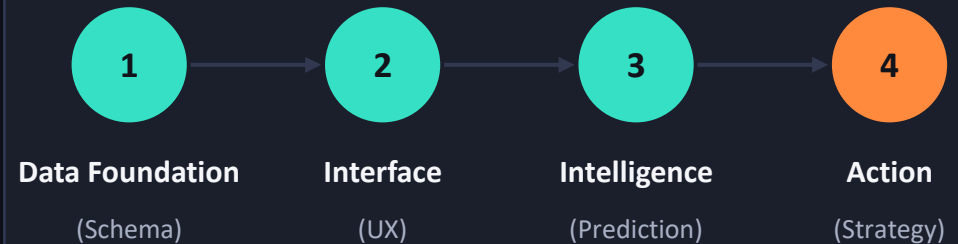
The Analytical Lens

The Siloed Reality

- Incident Data
- Location Data (10 facilities)
- Response Data
- Weather Data

850 raw records spanning independent systems, with no shared structure.

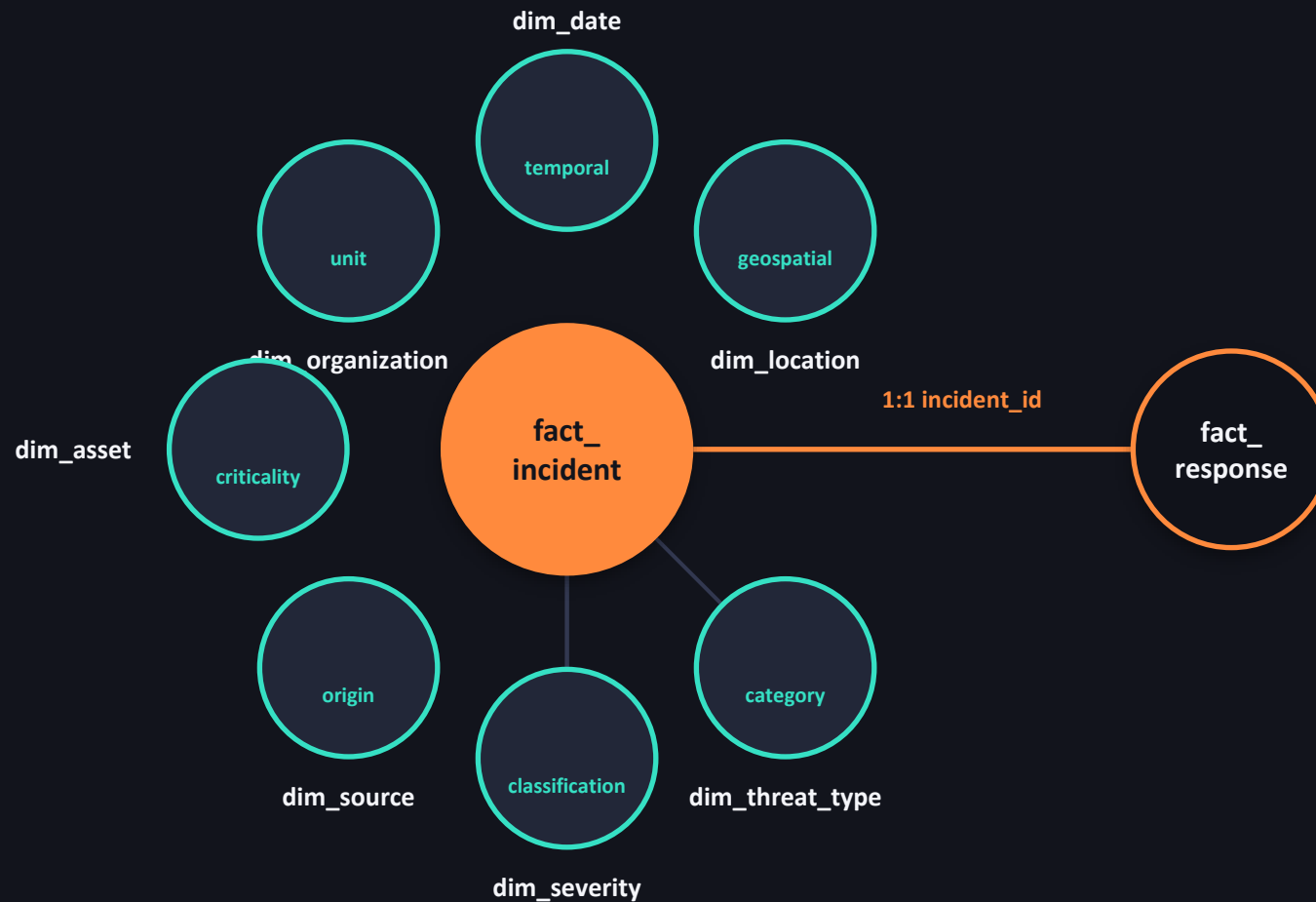
The Unified Vision



A single centralized engine capable of identifying emerging threats and optimizing resource allocation.

SHIFT Moving from distributed, unstandardized operational logs to a single centralized intelligence engine that identifies emerging threats and optimizes resource allocation.

Data Foundation & Architecture



Star Schema Summary

- One incident fact table, seven descriptive dimensions.
- A 1:1 relationship links every incident to its response record.
- Standardizes 8 distinct reporting sources into a single source of truth.
- Enables cross-filtering across 90 days of activity.

Baseline Operational KPIs

850

Total Incidents

89.76%

SLA Compliance

10.39 min

Avg Response Time

53.4

Baseline Threat Severity Index (Medium band)

67

Critical Incidents

316 / 37.2%

Open Incident Backlog

TAKEAWAY High incident volume with strong overall SLA compliance — but a 37.2% open backlog and 67 critical incidents represent immediate operational friction.

Interactive User Experience

1

Drill-Through

Navigates from a macro regional view directly into the City Details page for specific incident logs and response metrics.

2

Custom Tooltips

Reveals deep contextual data — asset criticality, precise weather conditions — on hover, without cluttering the main analytical view.

3

Cross-Filtering

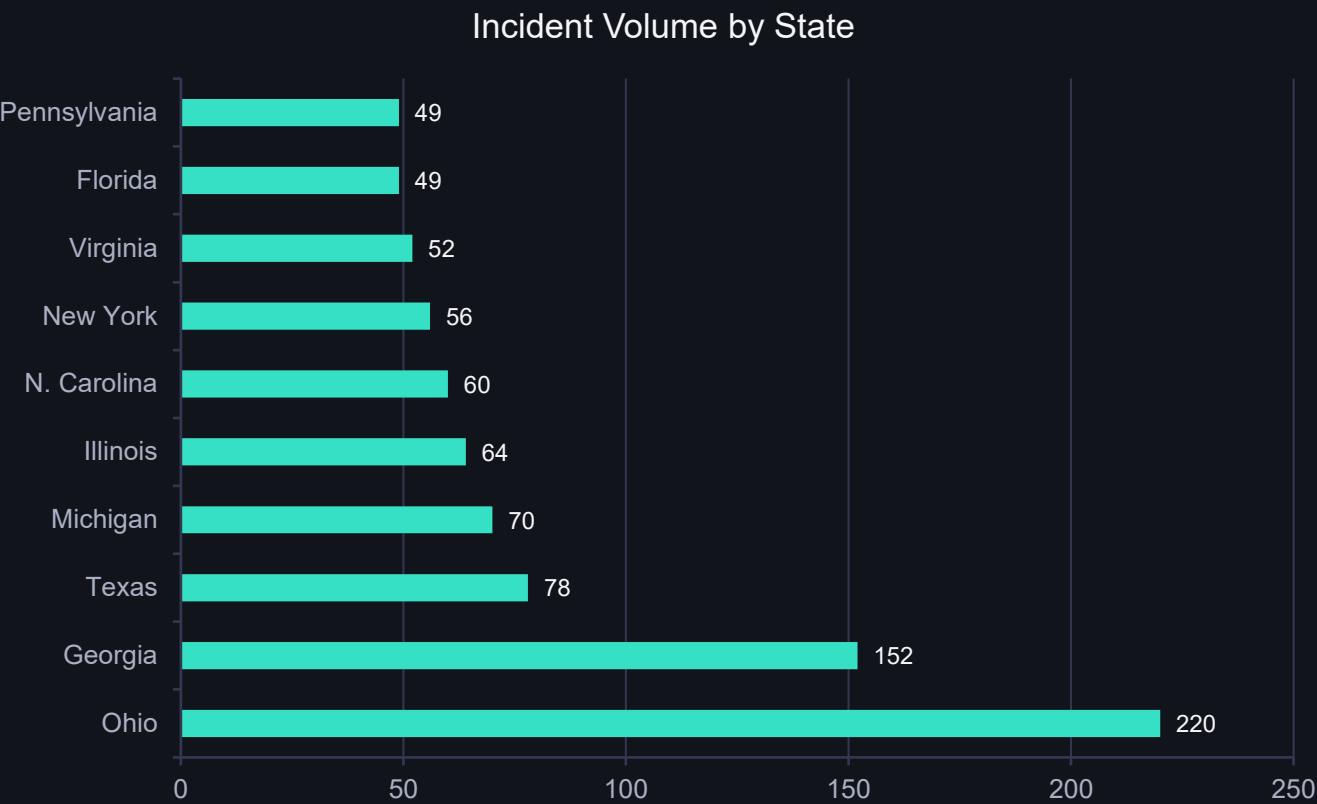
Selecting a threat category instantly updates:

Severity Metrics

Geospatial Maps

Temporal Trends

Geospatial Intelligence & Risk



43.8%

Ohio + Georgia account for 372 of 850 incidents

Cincinnati

Composite Risk 72.1

Avg Severity 53.44

Savannah

Composite Risk 71.1

Avg Severity 52.46

CORE INSIGHT Risk is driven by severity, not just volume. Cincinnati and Savannah carry the highest composite risk scores despite lower raw incident counts.

Temporal & Behavioral Analytics

1

Phase 1

The Trigger

467

After-Hours

383

Business Hours

2

Phase 2

The Anomaly Spikes

16 / day

vs. 9.4 average — 2.2 z-score

Spike window

Mar 15 → May 29

3

Phase 3

The Triage Scale

Response Resolve

Low Severity

7.61m → 110.2m

Critical Severity

14.17m → 154.5m

CONCLUSION Triage protocols function properly — severity dictates response effort — but the staffing model is heavily misaligned with the after-hours threat reality.

Security Risk Assessment

Regional Risk Scorecard

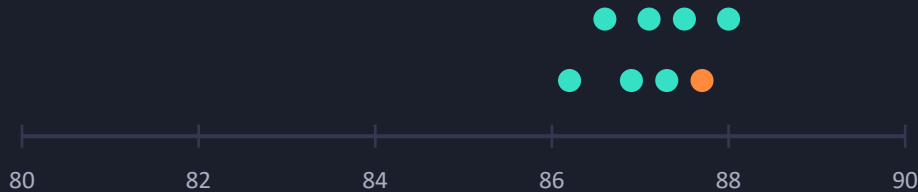
Overall Security Risk Score **81.36**

Northern District	81.63	Critical
Eastern Corridor	81.52	Critical
Central Hub	79.45	Elevated
Southern Reach	77.80	Elevated
Western Sector	76.91	Elevated

Asset Vulnerability Baseline

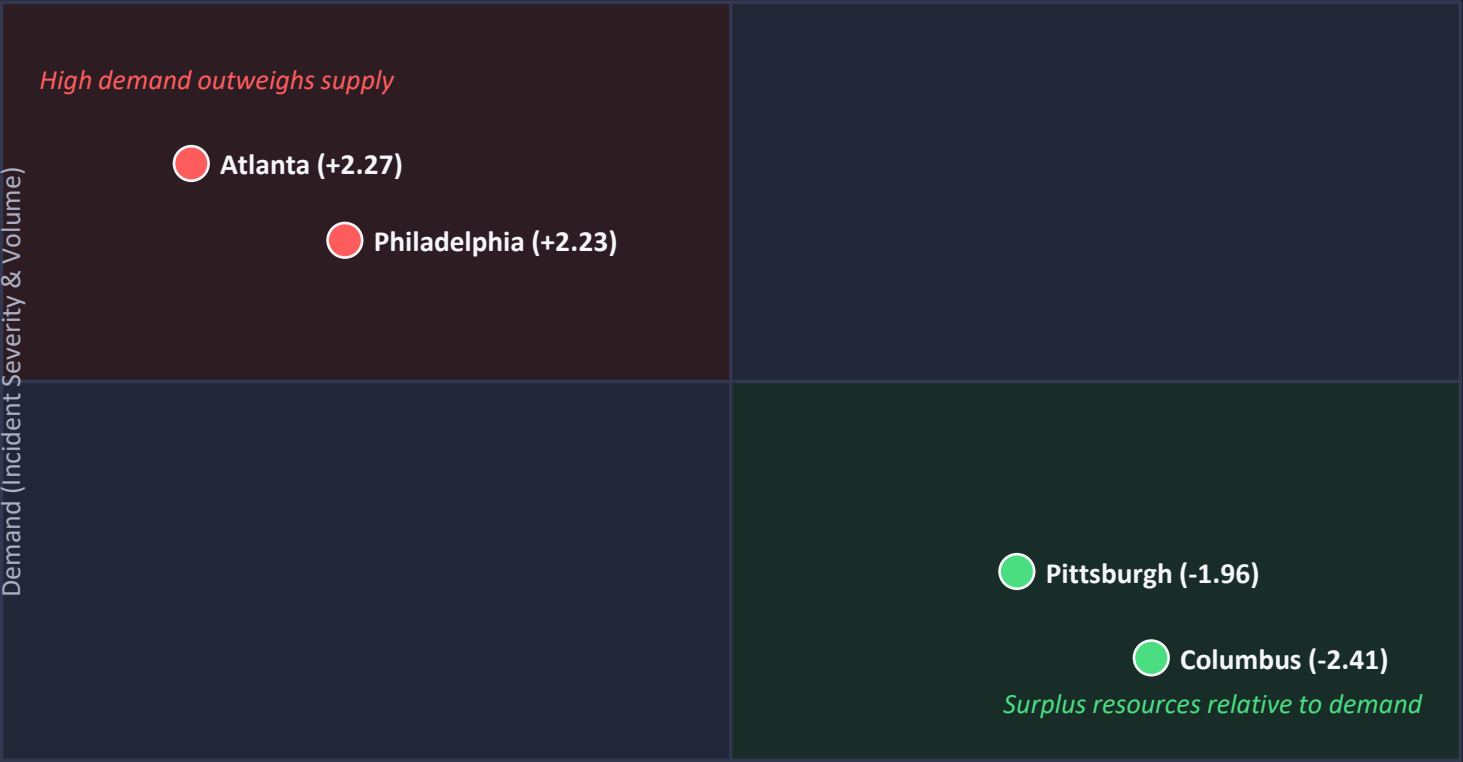
Vulnerability scores across all 8 asset types, plotted 80 → 90

Admin Office & Data Center — marginally highest



KEY INSIGHT Vulnerability is an organization-wide, systemic issue tightly clustered across all 8 asset types — mitigation requires systemic policy changes, not isolated spot-fixes.

Diagnostic: Resource Allocation Gap



Reading the Matrix

Top-left (red)

Cities where threat demand exceeds allocated response capacity — under-resourced.

Bottom-right (green)

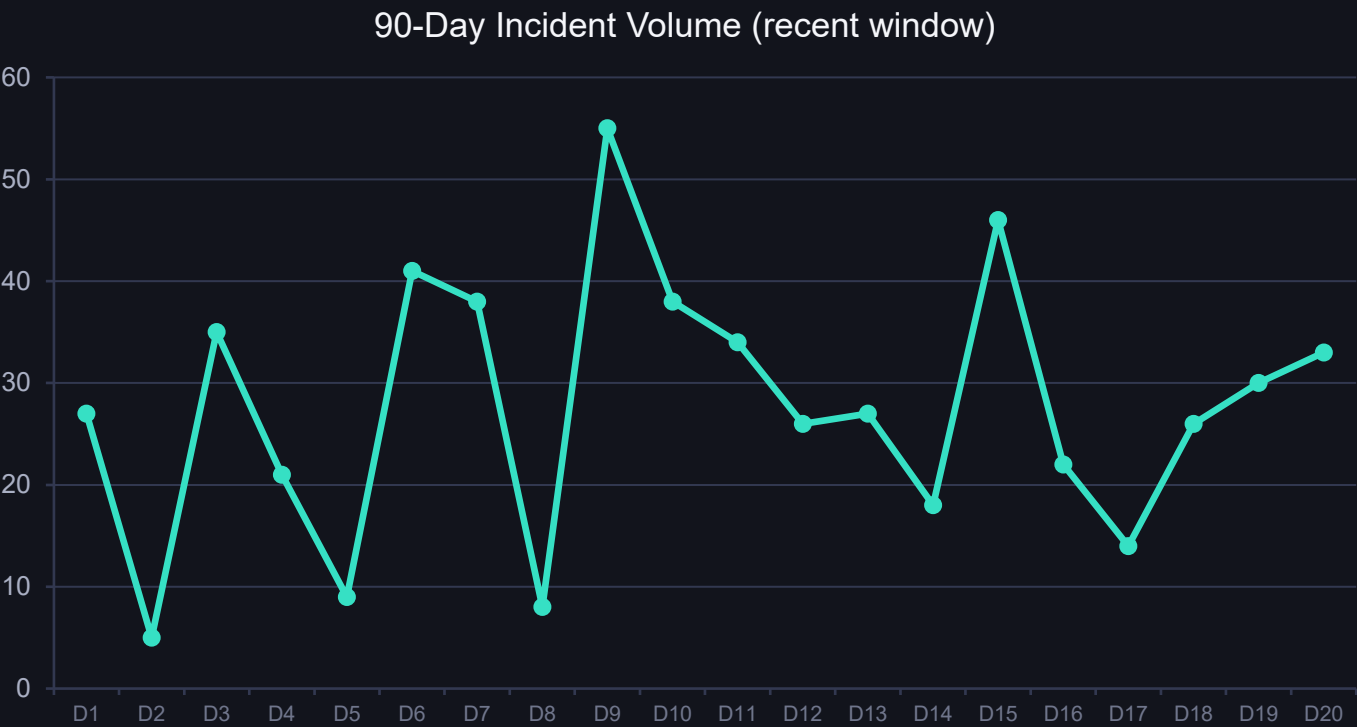
Cities with more allocated resources than current threat demand justifies.

Gap score

Standardized difference between demand and supply; larger magnitude = larger mismatch.

TAKEAWAY Resource allocation is measurably uneven. Operational efficiency requires immediate resource redistribution across these specific city pairs.

Predictive Intelligence & Forecasting



Forecast Accuracy (MAE)

2.71 incidents / day

14-day forward projection window

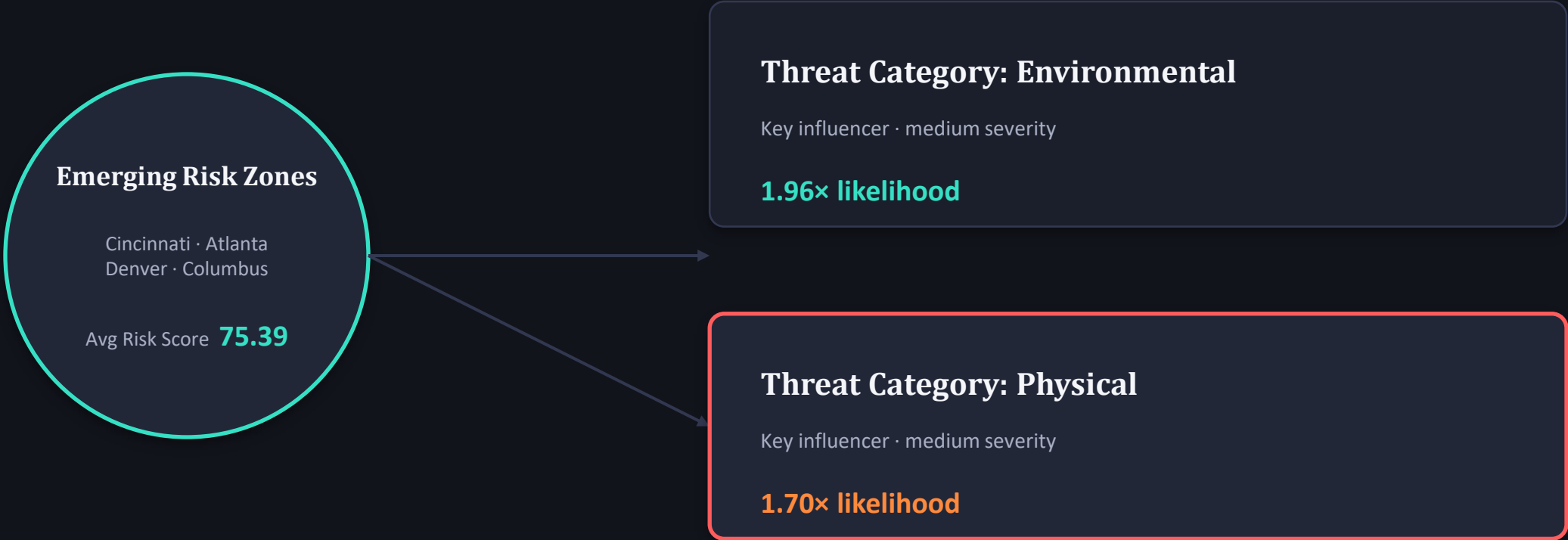
Predicted Risk (14-day horizon)

53.40

Confidence band widens toward the outer bound of the window

KEY INSIGHT The confidence band widens explicitly toward the end of the 14-day window — planning decisions should weight near-term forecasted days more heavily than the outer bounds.

Emerging Risk Zones & Influencers



CONVERGENCE NOTE Physical incidents are independently flagged by both the Key Influencers model and the Security Weakness module — a highly validated signal for intervention.

Key Insights Synthesis

1

Risk Defies Volume

Incident volume concentrates geographically in Ohio and Georgia, but composite risk is driven entirely by severity in Cincinnati and Savannah. A volume-only staffing model fails.

2

Predictable Responses, Shifting Threats

Average response time is remarkably uniform (10.0–10.8 min) regardless of geography. However, environmental incidents grew 8.1% in the second half of the period.

3

The Silent Backlog

37.2% of all incidents (316 total) remain open. This is not a throughput failure — SLA compliance stays high at 89.76% — but an investigation and escalation bottleneck.

Strategic Recommendations

1

Immediate Tactical Rebalance

Shift response resources from Columbus and Pittsburgh to Atlanta and Philadelphia based on identified gap scores. Rebalance daily shift schedules to cover the after-hours threat dominance.

2

Targeted Threat Mitigation

Prioritize physical security interventions specifically within the Critical-band Northern District and Eastern Corridor regions.

3

New Operational KPIs

Track the 316-incident open backlog as a standing operational metric. Revise the city-level risk banding logic so it stops reading 10 of 10 cities as “High Risk.”

FROM FRAGMENTED DATA TO STRATEGIC FORESIGHT

Four unstandardized, siloed data feeds have been transformed into an executive-ready, predictive analytical framework — achieving a validated 2.71 MAE forecast accuracy.

Leadership can now proactively allocate resources based on forecasted severity — rather than reacting retrospectively to raw incident volume.